# Bassem Krayem

**Backend Software Engineer · Node.js REST APIs & Python**

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## PROFILE

Backend Software Engineer and Holberton School graduate (Software Engineering Program), focused on designing clean, scalable REST APIs and backend systems. Comfortable owning a project end-to-end — from architecture and database design to deployment — and skilled at using modern AI-assisted development tools (GitHub Copilot) to move faster without sacrificing code quality. Currently building toward DevOps and Cloud Engineering, growing hands-on skills in CLI-driven automation, CI/CD, and infrastructure workflows. Thrives in collaborative, fast-paced environments and quick to pick up new tools and patterns.

## SKILLS

**Languages:** Node.js (JavaScript), Python, Bash
**Databases:** MongoDB (Mongoose), PostgreSQL, MySQL
**Backend & APIs:** Express.js, REST API Design, JWT Auth, Backend Architecture
**Tools:** Linux/Ubuntu, Git/GitHub, Mocha/Chai, Swagger/OpenAPI

## PROJECTS

**Motamyezon — Digital Books REST API**

- Built a production-style REST API for an Arabic digital book platform, with full CRUD for books, authors, series, and categories, supporting multi-format uploads (EPUB, AZW3, KFX).
- Implemented JWT authentication with role-based access control (user, uploader, admin), including signup, login, and secure password reset flows.
- Hardened the API against common attack vectors — NoSQL injection, XSS, parameter pollution — using helmet, express-mongo-sanitize, xss-clean, hpp, and rate limiting.
- Integrated DigitalOcean Spaces (S3-compatible) for cloud file storage and documented the API with Swagger/OpenAPI for interactive testing.
- Wrote an extensive Mocha/Chai test suite covering authentication, CRUD operations, permissions, and input validation across all resources.

**Stack:** Node.js, Express, MongoDB/Mongoose, JWT, Joi, Mocha/Chai, DigitalOcean Spaces (S3-compatible)
**Repo:** github.com/bassem-krayem/motamyezon_for_text_books_v2

**Markdown to HTML Parser** *Holberton School coursework*

- Built a Python CLI tool that parses core Markdown syntax (headings, lists, bold/emphasis, paragraphs) into valid HTML entirely from scratch, without any Markdown library.
- Handled custom syntax extensions and edge cases — including MD5-hash text conversion — with proper CLI argument validation and error handling.

**Stack:** Python, PEP8

## EDUCATION

**Holberton School — Software Engineering Program** *Feb 2024 – Jun 2026*

- Graduated from a rigorous project-based, peer-to-peer collaborative learning model.
- "Learning how to learn" methodology — solving complex engineering problems hands-on.
- Built strong foundations in industry-standard backend technologies and scalable architecture patterns.

## CERTIFICATIONS

- **Back End Development and APIs** — freeCodeCamp
- **Relational Database** — freeCodeCamp
- **Quality Assurance** — freeCodeCamp

## LANGUAGES

**Arabic:** Native **English:** Intermediate